

Validity Analysis: WXIMPACTBENCH

Target context: Argentine Government Disaster Reporting Reliability — Policy Staff

Overall risk: **HIGH**

Deployment Context

Use case: Use case: Reliability of disaster reporting in news at national/local level, finding alignment with effective and verified information (e.g. noisy vs real signals). The evaluation data could report or not disaster damage.

Target population: Policy makers and government agencies in Argentina studying trustiness and reliability of disaster communication across different channels are (e.g. news, social media post, press). They would be based in Buenos Aires, speakers of Rioplatense Spanish, with an intermediate to advanced level of English, at least a master's degree, and be part of an upper-middle-class household.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology ✓	3	Moderate gaps	high
Input Content △	1	Serious concern	high
Input Form △	2	Significant gaps	high
Output Ontology △	2	Significant gaps	high
Output Content △	2	Significant gaps	high
Output Form ✓	3	Moderate gaps	medium
Average	2.2		

△ = highest concern ✓ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

WXImpactBench is an English-language, formally-styled, North America-centric benchmark whose six top-level impact categories transfer reasonably to Argentine disaster types but whose content, register, language, annotator pool, and output schema all diverge substantially from the deployment's needs. The most severe validity violations are in `input_content` (zero Spanish-language, zero Argentine, zero social media content; HIGH priority) and `output_ontology` (no native credibility scoring — though deliberately handled downstream by user design — and no sub-labels for Argentine-specific granularity; HIGH

priority). Input_form is mismatched on register and length, and the paper itself acknowledges formal historical text is easier for LLMs than informal modern text, meaning benchmark scores likely overstate deployment-relevant capability. Output_content carries zero Argentine annotator representation and exhibits empirical labeling inconsistencies on positively-framed chunks. The benchmark may serve as one input to model triage but cannot stand alone as evidence of deployment-readiness for Argentine government disaster reporting reliability assessment.

Practical Guidance

What This Benchmark Measures

WXImpactBench measures LLM ability to extract six pre-defined societal impact categories from formal long-form English-language historical journalism, and to retrieve relevant articles given a generated question. For the Argentine deployment, this provides indirect evidence of LLM reasoning over formal press content, particularly across two linguistic registers, and exercises some noise-rejection capability through the chunked dataset's heterogeneous off-topic content [DATASET-D2, D7, D23]. Strongest dimensions for the target context are output_form (functional compatibility for upstream feature use) and input_ontology (six top-level categories applicable per elicitation [A1]).

Construct Depth

Construct depth is shallow for the deployment's primary need. The benchmark probes formal English long-form impact classification but does not probe (a) Spanish-language disaster content, (b) social media or short-form informal text, (c) credibility/reliability scoring, (d) Argentine institutional actor recognition, (e) slow-onset disaster framing (ENSO drought), or (f) villa miseria social vulnerability framing. The paper's own observation that formal historical style is easier for LLMs [Q83, Q85] caps the upper bound of construct generalization. Empirical labeling inconsistencies [DATASET-D17, D22, D27] further limit confidence in the convergent validity of the impact-classification construct itself.

What Else You Need

The deployment requires substantial supplementation: (1) a Rioplatense Spanish disaster impact gold-standard test set built with Argentine annotators (no equivalent exists per WEB-14, gap_id 7); (2) a social media short-form evaluation, with HumAID/CrisisBench [WEB-14, WEB-15] as English-language starting points pending cross-lingual transfer validation; (3) a credibility/reliability scoring layer with Argentine source metadata (outlet reach, account verification, AFE/SINAGIR institutional recognition); (4) sub-label refinement for villa miseria infrastructure, agricultural commodity-specific loss, and Argentine partisan-framing political impact; (5) re-annotation of a sample by Argentine emergency management experts to verify whether documented labeling inconsistencies persist under Argentine interpretation.

ID	Evidence
Q83	"The reason might be the structured and formal narrative style used to report disruptive weather events in historical periods, which more explicitly highlights cause-and-effect relationships." (p.8)
Q85	"Though the modern text might dominate within the pre-trained corpus, the language patterns used in historical narrative styles are easier for language models to identify, and thus perform better on classifying disruptive weather impacts." (p.8)
D2	But the real beneficiaries of the work-at-home trend are the employees themselves... She takes a turn at the table in a lay way (5) 29, A good one is of assistance (9) Down 1, Full of ill-will and disease (9)... — Non-weather content: work-from-home editorial + crossword puzzle clues
D7	National Lampoon's Van Wilder 2: The Rise of Taj Starring: Kal Penn, Lauren Cohan and Daniel Percival Playing at: AMC, Brossard, Cavendish, Colossus, LaSalle cinemas. — Film review; no weather content at all
D17	PETER MARTIN, GAZETTE Workers from the Connecticut Light and Power Co were working to reconnect power lines on Oxford Ave in Notre Dame de Grace... Temperatures plummet; today's high minus-15C... more than one million people struggle on without power or heat — 1998 Quebec ice storm coverage describing power outages and extreme cold; labeled all-zero — potential annotation gap
D22	Almost 4 million people across China had been cut off by flood waters, 810,000 homes have collapsed and 2.8 million homes have been damaged in eight provinces... Japan struck at young and old yesterday, killing a schoolgirl and an 85-year-old woman — 1996 China Yangtze floods + Japan E. coli outbreak in same chunk; infrastructural=1 but human_health=0 despite explicit casualties
D23	-2 City res y 5000 5 5 5 Aiexavn 8000 50 50 50 Ckearcdnf 835 825 25..." / "Escenvt 115780 298 J80 289 19 Caiunoil 8000 J9 J9 29... — OCR-corrupted stock market tables; repeated for same article id across multiple rows

D27	A poor lady passenger was dashed to leeward and had her skull fractured. She was landed this morning in a dying state with a coffined child that succumbed last night to its sufferings. — 1881 Atlantic storm with explicit casualties (skull fracture, child death) labeled all-zero
WEB-14	No complementary Argentine Spanish-language benchmark exists for cross-validation (https://crisisnlp.qcri.org/humaid_dataset)
WEB-15	https://ojs.aaai.org/index.php/ICWSM/article/download/18116/17919/21611

Input Ontology (IO)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: WXIMPACTBENCH | Context: Argentine Government Disaster Reporting Reliability — Policy Staff

Input Ontology definition: Whether the benchmark’s test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The six top-level impact categories (Infrastructural, Political, Financial, Ecological, Agricultural, Human Health) are broadly applicable to Argentine disaster types and the elicitation confirms they 'could accommodate the named disaster types' [A1]. Empirical inspection confirms each category appears in the data with positive examples [DATASET-D14, D16, D18]. However, the ontology is constructed entirely around formal newspaper articles [Q18, Q115], with no task type targeting social media, press releases, or short-form informal content — the deployment's highest-priority input channels. No sub-labels exist within the six categories, a recognized gap for Argentine policy granularity (e.g., villa miseria vs. formal infrastructure). The chunked dataset also includes a high proportion (~65%) of off-topic zero-label fragments [DATASET-D1, D2, D7, D23] that may dilute the discriminative validity of the ontology for genuine disaster classification. The QA task is closed-retrieval only [Q89], precluding the open-retrieval setting most relevant to a deployment scanning real source streams.

ID	Evidence
Q18	"Climate impact analysis (Thulke et al., 2024) aims to help society make correct decisions about climate-related challenges affecting communities, e.g., understanding the weather impacts on society." (p.2)
Q89	"The practical open-retrieval setting, i.e., identifying the relevant articles from a huge database, is left for future studies, which could further facilitate knowledge enhancement in understanding disruptive weather impacts." (p.9)
Q115	"Table 16 presents the prompt designed to analyze historical newspaper texts and classify them into six distinct impact categories based on explicit mentions of weather-related events." (p.16)
D1	John's Cloudy -4 -4 United States Max Min Atlanta Cloudy 17 4 Boston Cloudy 3 -6 Chicago Cloudy 2 2 Dallas Thunderstorms 11 2... — Weather forecast table with temperature/condition data; no impact content
D2	But the real beneficiaries of the work-at-home trend are the employees themselves... She takes a turn at the table in a lay way (5) 29, A good one is of assistance (9) Down 1, Full of ill-will and disease (9)... — Non-weather content: work-from-home editorial + crossword puzzle clues
D7	National Lampoon's Van Wilder 2: The Rise of Taj Starring: Kal Penn, Lauren Cohan and Daniel Percival Playing at: AMC, Brossard, Cavendish, Colossus, LaSalle cinemas. — Film review; no weather content at all
D14	by six o'clock the electric car services in all the important points west of Toronto had been completely paralyzed... an unprecedented snowfalls and heavy winds. — 1894 blizzard; infrastructure (transit) disruption labeled; North American geography
D16	Aid Laurent said that there were mountains of snow all over the city. Already \$30,000 has been expended in removing it... Seven hundred and fifty men and 450 carters were employed... — 1887 Montreal snow removal debate in city council; all three labeled labels are plausible
D18	more than 100,000 homes damaged and crops on about 175,000 hectares of farmland destroyed... death toll in Chongqing in China's southwest rose to 42 people and 12 missing from torrential downpours — 2007 global flood roundup (England, China, Pakistan) — non-Canadian events used as positive examples
D23	-2 City res y 5000 5 5 5 Aiexavn 8000 50 50 50 Ckearcdnf 835 825 25..." / "Escenvt 115780 298 J80 289 19 Caiunoil 8000 J9 J9 29... — OCR-corrupted stock market tables; repeated for same article id across multiple rows

Strengths

- Six top-level categories map cleanly onto Argentine disaster impact types per elicitation [A1, Q11]
- All six labels are empirically present and distinguishable in the data, with both single-label and multi-label positive cases [DATASET-D14, D16, D18]

- Multi-label simultaneous classification design [Q57] matches realistic disaster reports that span multiple impact dimensions

ID	Evidence
Q11	"The articles are selected by topic modeling, including six impact categories (infrastructural, political, financial, ecological, agricultural, and human health), which are informed by previous studies (Imran et al., 2016a) and align with modern disaster impact assessment frameworks (Silva et al., 2022)." (p.2)
Q57	"Different from traditional methods that decompose multi-label text classification into multiple binary classification tasks (Boutell et al., 2004; Liu et al., 2017), we simultaneously identify all relevant disruptive weather impacts for each input by calling the LLM once." (p.6)
D14	by six o'clock the electric car services in all the important points west of Toronto had been completely paralyzed... an unprecedented snowfalls and heavy winds. — 1894 blizzard; infrastructure (transit) disruption labeled; North American geography
D16	Aid Laurent said that there were mountains of snow all over the city. Already \$30,000 has been expended in removing it... Seven hundred and fifty men and 450 carters were employed... — 1887 Montreal snow removal debate in city council; all three labeled labels are plausible
D18	more than 100,000 homes damaged and crops on about 175,000 hectares of farmland destroyed... death toll in Chongqing in China's southwest rose to 42 people and 12 missing from torrential downpours — 2007 global flood roundup (England, China, Pakistan) — non-Canadian events used as positive examples

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Argentine deployment requires categories covering: urban flash floods (AMBA), sudestada coastal flooding, pampero windstorms, ENSO-driven Pampa drought, Patagonian wildfire, NOA flood/drought, plus credibility/reliability scoring across formal press, social media, and informal community channels [WEB-1, WEB-5, WEB-7].

IO-2: Benchmark omits: (a) any task targeting social media or short-form informal content [Q18, Q115], (b) credibility/misinformation/source-reliability scoring (the deployment's core task per A3), (c) sub-labels within the six categories needed for Argentine policy granularity [A1], (d) open-retrieval QA setting [Q89], and (e) Spanish-language task framing.

IO-3: No categories appear actively irrelevant to the Argentine context — the six top-level categories are broadly applicable per A1. However, the LDA-derived 15 weather event types [Q101] are not surfaced as evaluation categories, and the chunked dataset contains a high proportion of off-topic content (crosswords, film reviews, stock tables) labeled all-zero [DATASET-D2, D7, D23], which dilutes the ontology's discriminative signal.

IO-4: Major gaps: (1) no social media / short-form input task type despite this being a HIGH-priority deployment channel; (2) no credibility scoring task; (3) no sub-labels for Argentine-specific granularity (informal settlements, agricultural commodity-specific loss); (4) closed-retrieval QA only.

ID	Evidence
Q18	"Climate impact analysis (Thulke et al., 2024) aims to help society make correct decisions about climate-related challenges affecting communities, e.g., understanding the weather impacts on society." (p.2)
Q89	"The practical open-retrieval setting, i.e., identifying the relevant articles from a huge database, is left for future studies, which could further facilitate knowledge enhancement in understanding disruptive weather impacts." (p.9)
Q101	"Using Latent Dirichlet Allocation, the dataset was categorized into 15 primary weather event types." (p.14)
Q115	"Table 16 presents the prompt designed to analyze historical newspaper texts and classify them into six distinct impact categories based on explicit mentions of weather-related events." (p.16)

D2	But the real beneficiaries of the work-at-home trend are the employees themselves... She takes a turn at the table in a lay way (5) 29, A good one is of assistance (9) Down 1, Full of ill-will and disease (9)... — Non-weather content: work-from-home editorial + crossword puzzle clues
D7	National Lampoon's Van Wilder 2: The Rise of Taj Starring: Kal Penn, Lauren Cohan and Daniel Percival Playing at: AMC, Brossard, Cavendish, Colossus, LaSalle cinemas. — Film review; no weather content at all
D23	-2 City res y 5000 5 5 5 Aiexavn 8000 50 50 50 Ckearcdnf 835 825 25..." / "Escenvt 115780 298 J80 289 19 Caiunoil 8000 J9 J9 29... — OCR-corrupted stock market tables; repeated for same article id across multiple rows
WEB-1	SINAGIR/AFE Argentine institutional framework operates with different political accountability structure than Canadian framework (https://www.argentina.gob.ar/sinagir/institucional)
WEB-5	98.4% Argentine mobile broadband indicates short-form mobile content dominant in deployment (https://datareportal.com/reports/digital-2026-argentina)
WEB-7	Rural-urban digital divide means disaster-zone content carries different noise characteristics than urban Buenos Aires (https://freedomhouse.org/country/argentina/freedom-net/2024)

Information Gaps

- Whether sub-label schema for villa miseria infrastructure damage could be added without re-annotating the corpus
- Whether credibility scoring could be derived from existing label outputs without dedicated task design

Requires Expert Verification

- Which Argentine sub-categories within each top-level label matter most for SINAGIR/AFE policy decision-making

Remediation

Gap 1: No task type for social media or short-form informal content; closed-retrieval QA only [Q18, Q89]

Recommendation: Augment evaluation with an open-retrieval QA task and a short-form social-media classification task using Argentine Twitter/X disaster content. Treat WXImpactBench's closed-retrieval scores as an upper-bound proxy and supplement with open-retrieval evaluation against an Argentine source corpus.

ID	Evidence
Q18	"Climate impact analysis (Thulke et al., 2024) aims to help society make correct decisions about climate-related challenges affecting communities, e.g., understanding the weather impacts on society." (p.2)
Q89	"The practical open-retrieval setting, i.e., identifying the relevant articles from a huge database, is left for future studies, which could further facilitate knowledge enhancement in understanding disruptive weather impacts." (p.9)

Input Content (IC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: WXIMPACTBENCH | Context: Argentine Government Disaster Reporting Reliability — Policy Staff

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark corpus is exclusively English-language historical Canadian newspaper text from La Presse, La Patrie, and the Montreal Gazette [Q97]. No Spanish-language, Argentine, or Latin American source material is represented, and no Argentine institutional actors (SINAGIR, SMN, AFE), geographic references, or local disaster framing appear. The deployment explicitly handles Rioplatense Spanish across formal press, social media, and informal community channels — none of which are in the benchmark. The paper itself acknowledges 'potential biases in underrepresented historical events and linguistic variations' [Q90] and that 'the interpretation of weather-related disruptions in historical newspapers might be influenced by demographic and contextual factors' [Q96]. Empirically, the dataset contains some non-Canadian wire-service content (China, Mediterranean, Indonesia) [DATASET-D18, D19, D21], which marginally broadens geographic scope but does not include South America or Spanish-language content. Web search confirmed no Argentine Spanish-language disaster NLP benchmark exists [WEB-14, WEB-15] to supplement this gap. Given that input_content is a HIGH-priority dimension per elicitation, this represents a major validity violation.

ID	Evidence
Q90	"Although WXIMPACTBENCH provides valuable insights (e.g., exhibit the strengths and weaknesses of various society impact understanding) about evaluating LLMs on disruptive weather, it may have potential biases in underrepresented historical events and linguistic variations." (p.9)
Q96	"The interpretation of weather-related disruptions in historical newspapers might be influenced by demographic and contextual factors, which is similar to other text datasets generated through crowd-sourcing with inherent challenges in ensuring that dataset labels are fully representative of diverse societal perspectives (Talat et al., 2022)." (p.9)
Q97	"Our primary data source is a corpus of three digitized newspapers (La Presse, La Patrie and Montreal Gazette), obtained through collaboration with the McGill University Library and Archives and the Bibliothèque nationale du Québec." (p.9)
D18	more than 100,000 homes damaged and crops on about 175,000 hectares of farmland destroyed... death toll in Chongqing in China's southwest rose to 42 people and 12 missing from torrential downpours — 2007 global flood roundup (England, China, Pakistan) — non-Canadian events used as positive examples
D19	A line of freak thunderstorms rumbling across the northeastern Mediterranean coast from Spain and southern France to the principality of Monaco brought two days of disastrous flash flooding... Raging waters uprooted trees and swept away cars as river levels rose by as much as 3 metres — Mediterranean flooding; ecological and political labels assigned; non-Canadian geography
D21	hundreds of houses, schools and other buildings sustained damage when one of the twin peaks of the volcano exploded. At least 20 people sustained minor injuries... The most powerful earthquake to strike western Washington state in 30 years injured four people — 1999 global disaster roundup (Indonesia, Washington state, NZ, Romania); labeled positive
WEB-14	No complementary Argentine Spanish-language benchmark exists for cross-validation (https://crisisnlp.qcri.org/humaid_dataset)
WEB-15	https://ojs.aaai.org/index.php/ICWSM/article/download/18116/17919/21611

Strengths

- Some non-Canadian wire-service content (China, Pakistan, Mediterranean, Indonesia) appears in positive examples [DATASET-D18, D19, D21], indicating impact categories are applied beyond strictly Canadian events
- Two temporal periods covered to introduce linguistic variation [Q13]
- Authors explicitly acknowledge potential biases in underrepresented events [Q90]

ID	Evidence
----	----------

Q13	"We extract a sample of articles from two time periods, which cover linguistic and social changes across different eras and increase linguistic complexity due to the different descriptions of weather events in different times (Campbell, 2013)." (p.2)
Q90	"Although WXIMPACTBENCH provides valuable insights (e.g., exhibit the strengths and weaknesses of various society impact understanding) about evaluating LLMs on disruptive weather, it may have potential biases in underrepresented historical events and linguistic variations." (p.9)
D18	more than 100,000 homes damaged and crops on about 175,000 hectares of farmland destroyed... death toll in Chongqing in China's southwest rose to 42 people and 12 missing from torrential downpours — 2007 global flood roundup (England, China, Pakistan) — non-Canadian events used as positive examples
D19	A line of freak thunderstorms rumbling across the northeastern Mediterranean coast from Spain and southern France to the principality of Monaco brought two days of disastrous flash flooding... Raging waters uprooted trees and swept away cars as river levels rose by as much as 3 metres — Mediterranean flooding; ecological and political labels assigned; non-Canadian geography
D21	hundreds of houses, schools and other buildings sustained damage when one of the twin peaks of the volcano exploded. At least 20 people sustained minor injuries... The most powerful earthquake to strike western Washington state in 30 years injured four people — 1999 global disaster roundup (Indonesia, Washington state, NZ, Romania); labeled positive

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Argentine deployment requires Rioplatense Spanish proficiency, recognition of Argentine institutional actors (SINAGIR, AFE, SMN, Prefectura Naval, provincial civil defense), and familiarity with local disaster framing including villa miseria vulnerability — none of which appears in benchmark inputs [Q97].

IC-2: Cultural alignment is poor: benchmark content is North American historical journalism [Q97]; elicitation flags that 'informal or social media data carries meaning and communication patterns that are likely not represented in the Canadian-sourced benchmark' [A2].

IC-3: Inputs require Western-specific knowledge: 19th-century Canadian institutional context (Q12, Q14), Montreal civic geography [DATASET-D16], and English-language journalism conventions. None transfers to Spanish-language Argentine social media or AMBA disaster reporting.

IC-4: INSUFFICIENT DOCUMENTATION on regional annotator recruitment for Argentine context; no Argentine annotators were involved per Q110 (annotators specialize in regional Canadian historical climate).

IC-5: Major content issues: zero Spanish-language content; zero Argentine geography; zero villa miseria framing; zero social media content; zero institutional actors relevant to AFE/SINAGIR. Confirmed via web search — no complementary Argentine Spanish disaster benchmark exists [WEB-14, search gap_id 2 and 7].

ID	Evidence
Q12	"We design a four-stage data construction pipeline that begins with a disruptive weather impact dataset in which we correct OCR errors in digitalized newspaper article extraction." (p.2)
Q14	"Historical newspapers often employed more descriptive and elaborate narratives compared to modern reporting styles (Bingham, 2010). These narratives frequently included outdated terminology, spelling variations, and evolving writing conventions (Campbell, 2013)." (p.2)
Q90	"Although WXIMPACTBENCH provides valuable insights (e.g., exhibit the strengths and weaknesses of various society impact understanding) about evaluating LLMs on disruptive weather, it may have potential biases in underrepresented historical events and linguistic variations." (p.9)

Q97	"Our primary data source is a corpus of three digitized newspapers (La Presse, La Patrie and Montreal Gazette), obtained through collaboration with the McGill University Library and Archives and the Bibliothèque nationale du Québec." (p.9)
Q110	"The annotation process was conducted by members of a research group specializing in uncovering the history of a region's climate change through the regional historical weather records." (p.15)
D16	Aid Laurent said that there were mountains of snow all over the city. Already \$30,000 has been expended in removing it... Seven hundred and fifty men and 450 carters were employed... — 1887 Montreal snow removal debate in city council; all three labeled labels are plausible
WEB-1	SINAGIR/AFE Argentine institutional framework operates with different political accountability structure than Canadian framework (https://www.argentina.gob.ar/sinagir/institucional)
WEB-5	98.4% Argentine mobile broadband indicates short-form mobile content dominant in deployment (https://datareportal.com/reports/digital-2026-argentina)
WEB-14	No complementary Argentine Spanish-language benchmark exists for cross-validation (https://crisisnlp.qcri.org/humaid_dataset)

Information Gaps

- Whether HumAID/CrisisBench English-language tweets could serve as partial supplements after cross-lingual transfer validation [WEB-14]
- Empirical performance of LLMs on Rioplatense Spanish disaster content (no Argentine benchmark exists for direct measurement)

Requires Expert Verification

- Whether formal Argentine press content (Clarín, Infobae, La Nación) closely follows transferable journalistic patterns as A2 hypothesizes
- Construction of an Argentine Spanish gold-standard test set for cross-lingual validation

Remediation

Gap 1: Zero Spanish-language, Argentine, or social media content in benchmark corpus [Q97, WEB-14]

Recommendation: Construct an Argentine Spanish disaster impact test set drawing from Clarín, Infobae, La Nación, government press releases, and Twitter/X content covering AMBA flooding, sudestada, Pampa drought, and Patagonian wind events. Use Argentine emergency management experts as annotators. Use HumAID/CrisisBench as methodological reference (not direct content reference) given the English-language gap.

ID	Evidence
Q97	"Our primary data source is a corpus of three digitized newspapers (La Presse, La Patrie and Montreal Gazette), obtained through collaboration with the McGill University Library and Archives and the Bibliothèque nationale du Québec." (p.9)
WEB-14	No complementary Argentine Spanish-language benchmark exists for cross-validation (https://crisisnlp.qcri.org/humaid_dataset)

Input Form (IF)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: WXIMPACTBENCH | Context: Argentine Government Disaster Reporting Reliability — Policy Staff

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark is text-only **[Q19]**, which matches the deployment's text-based channels. However, the form characteristics diverge substantially: benchmark inputs are long-form (avg 2,987 tokens) or chunked (avg 781 tokens) **[Q109]**, formally structured historical journalism with 'outdated terminology, spelling variations, and evolving writing conventions' **[Q14]**. The Argentine deployment includes social media posts (often <280 chars), WhatsApp messages, hashtags, emoji, lunfardo slang, and informal abbreviation — none of which is represented. Critically, the paper itself notes that formal historical narrative style is *easier* for LLMs to classify **[Q83, Q85]**, meaning benchmark performance estimates likely overstate model capability on noisier informal inputs. Web research confirms most LLMs face specific challenges with flood-related social media even in English **[WEB-17]**. The benchmark is English-only with Latin script; Rioplatense Spanish, while also Latin-script, introduces voseo, yeísmo rehilado lexicon, and Italian-influenced vocabulary not exercised. Empirically, the dataset contains heavily OCR-corrupted financial table chunks **[DATASET-D23]** which are not analogous to Argentine social media noise but do confirm noise tolerance is partially exercised. Input form is a MODERATE-priority dimension per elicitation; score reflects significant register and length mismatch.

ID	Evidence
Q14	"Historical newspapers often employed more descriptive and elaborate narratives compared to modern reporting styles (Bingham, 2010). These narratives frequently included outdated terminology, spelling variations, and evolving writing conventions (Campbell, 2013)." (p.2)
Q19	"The construction of the dataset aims to obtain high-quality text samples. The pipeline overview is presented in Figure 2, which consists of four stages: data collection, post-OCR correction, topic-aware article selection, and manual label annotation." (p.3)
Q83	"The reason might be the structured and formal narrative style used to report disruptive weather events in historical periods, which more explicitly highlights cause-and-effect relationships." (p.8)
Q85	"Though the modern text might dominate within the pre-trained corpus, the language patterns used in historical narrative styles are easier for language models to identify, and thus perform better on classifying disruptive weather impacts." (p.8)
Q109	"The average number of tokens per article is 2987.4 in long-context settings and 781.3 in mixed-context settings." (p.15)
D23	-2 City res y 5000 5 5 5 Aiexavn 8000 50 50 50 Ckearcdnf 835 825 25..." / "Escenvt 115780 298 J80 289 19 Caiunoil 8000 J9 J9 29... — OCR-corrupted stock market tables; repeated for same article id across multiple rows
WEB-17	Most LLMs face specific challenges processing flood-related social media data even in English (HumAID-based six-LLM analysis) (https://www.semanticscholar.org/paper/HumAID:-Human-Annotated-Disaster-Incidents-Data-Alam-Qazi/462cc2046ef4d48d844813b66d8a1ed6dfda3bc0)

Strengths

- Text modality matches deployment's text channels **[Q19]**
- Mixed-context chunking (avg 781 tokens) **[Q109]** is closer to press release length than full historical articles
- Two linguistic registers (historical and modern) exercise some register robustness **[DATASET-D14, D18]**
- OCR-noisy chunks in dataset **[DATASET-D23]** provide some test of noise tolerance

ID	Evidence
Q19	"The construction of the dataset aims to obtain high-quality text samples. The pipeline overview is presented in Figure 2, which consists of four stages: data collection, post-OCR correction, topic-aware article selection, and manual label annotation." (p.3)

Q109	"The average number of tokens per article is 2987.4 in long-context settings and 781.3 in mixed-context settings." (p.15)
D14	by six o'clock the electric car services in all the important points west of Toronto had been completely paralyzed... an unprecedented snowfalls and heavy winds. — 1894 blizzard; infrastructure (transit) disruption labeled; North American geography
D18	more than 100,000 homes damaged and crops on about 175,000 hectares of farmland destroyed... death toll in Chongqing in China's southwest rose to 42 people and 12 missing from torrential downpours — 2007 global flood roundup (England, China, Pakistan) — non-Canadian events used as positive examples
D23	-2 City res y 5000 5 5 5 Aiexavn 8000 50 50 50 Ckearcdnf 835 825 25..." / "Escenvt 115780 298 J80 289 19 Caiunoil 8000 J9 J9 29... — OCR-corrupted stock market tables; repeated for same article id across multiple rows

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Significant signal mismatch: benchmark long-context avg 2,987 tokens, mixed-context avg 781 tokens [Q109]; deployment includes tweet-length (<280 chars), WhatsApp messages, and government press bulletins. Register mismatch: formal English journalism vs. Rioplatense Spanish informal social media with lunfardo, emoji, hashtags.

IF-2: Argentine infrastructure supports text capture: 90.6% national internet penetration, 98.4% mobile broadband [WEB-5], Buenos Aires fixed broadband 41–101 Mbps [WEB-7]. However, rural disaster-zone producers have lower connectivity (three provinces <20 Mbps) [WEB-7, WEB-8], potentially yielding compressed/delayed content not represented in benchmark.

IF-3: Domain-specific form differences: deployment requires handling Twitter/X short-form, WhatsApp messages, emoji, hashtags, code-switching English/Spanish, OCR-absent native digital text — none in benchmark [Q19, Q109].

IF-4: External validity harm is significant: paper explicitly notes formal historical style is easier for LLMs [Q83, Q85], and HumAID-based research shows LLMs struggle with flood social media even in English [WEB-17]. Performance metrics on this benchmark likely overstate Argentine deployment capability on informal channels.

ID	Evidence
Q19	"The construction of the dataset aims to obtain high-quality text samples. The pipeline overview is presented in Figure 2, which consists of four stages: data collection, post-OCR correction, topic-aware article selection, and manual label annotation." (p.3)
Q83	"The reason might be the structured and formal narrative style used to report disruptive weather events in historical periods, which more explicitly highlights cause-and-effect relationships." (p.8)
Q85	"Though the modern text might dominate within the pre-trained corpus, the language patterns used in historical narrative styles are easier for language models to identify, and thus perform better on classifying disruptive weather impacts." (p.8)
Q109	"The average number of tokens per article is 2987.4 in long-context settings and 781.3 in mixed-context settings." (p.15)
WEB-5	98.4% Argentine mobile broadband indicates short-form mobile content dominant in deployment (https://datareportal.com/reports/digital-2026-argentina)
WEB-7	Rural-urban digital divide means disaster-zone content carries different noise characteristics than urban Buenos Aires (https://freedomhouse.org/country/argentina/freedom-net/2024)
WEB-8	https://ts2.tech/en/state-of-internet-access-in-argentina-fiber-5g-and-satellite-in-2025/
WEB-17	Most LLMs face specific challenges processing flood-related social media data even in English (HumAID-based six-LLM analysis) (https://www.semanticscholar.org/paper/HumAID:-Human-Annotated-Disaster-Incidents-Data-Alam-Qazi/462cc2046ef4d48d844813b66d8a1ed6dfda3bc0)

Information Gaps

- Empirical token-length distribution of Argentine disaster reporting across formal press, press releases, and social media
- Quantitative LLM performance gap between formal long-form and informal short-form Spanish disaster content

Requires Expert Verification

- Whether the mixed-context (~781 token) chunks adequately approximate Argentine government press bulletin length

Remediation

Gap 1: Long-form formal register only; deployment includes short-form informal Spanish social media [Q83, Q109, WEB-17]

Recommendation: Run a parallel evaluation on Spanish-translated tweet-length disaster content and on Argentine government press bulletins (~100–300 tokens) to quantify performance degradation relative to formal long-form benchmarks. Calibrate downstream credibility scoring to discount benchmark performance on informal channels by the measured gap.

ID	Evidence
Q83	“The reason might be the structured and formal narrative style used to report disruptive weather events in historical periods, which more explicitly highlights cause-and-effect relationships.” (p.8)
Q109	“The average number of tokens per article is 2987.4 in long-context settings and 781.3 in mixed-context settings.” (p.15)
WEB-17	Most LLMs face specific challenges processing flood-related social media data even in English (HumAID-based six-LLM analysis) (https://www.semanticscholar.org/paper/HumAID:-Human-Annotated-Disaster-Incidents-Data-Alam-Qazi/462cc2046ef4d48d844813b66d8a1ed6dfda3bc0)

Output Ontology (OO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: WXIMPACTBENCH | Context: Argentine Government Disaster Reporting Reliability — Policy Staff

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The deployment's primary decision-support output — reliability/credibility scoring of disaster reports — is structurally absent from the benchmark output space, which produces only six binary impact labels [\[Q25, Q33, Q117\]](#) and ranking lists [\[Q36\]](#). The user envisions credibility as a downstream layer aggregating model outputs with external metadata [\[A3\]](#), so this gap is partially mitigated by deployment design. However, even within the impact-labeling output, the binary schema with no sub-labels limits granularity for Argentine policy needs (villa miseria vs. formal infrastructure; soybean/maize vs. Canadian grain framing). Annotation guidelines emphasize 'direct and immediate effects' and 'explicit references' [\[Q118, Q147\]](#), excluding inferred or slow-onset impacts — a meaningful constraint for ENSO-driven Pampa drought assessment. Empirically, dataset analysis identifies labeling inconsistencies on positive-content chunks (1998 Quebec ice storm with 1M+ without power labeled all-zero [\[DATASET-D17\]](#); 1881 storm with explicit casualties labeled human_health=0 [\[DATASET-D27\]](#)) that suggest the binary output decisions may not be fully reliable. Output ontology is HIGH-priority per elicitation; score reflects the structural absence of the credibility dimension and granularity constraints, partially offset by user's deliberate design to handle credibility downstream.

ID	Evidence
Q25	"Six vulnerability-related disruptive weather impacts are defined as the labeling categories, including Infrastructural, Political, Financial, Ecological, Agricultural, and Human Health." (p.4)
Q33	"Specifically, given an article sample x_t corresponding to six ground-truth impacts $Y_t = \{y^i_t\}_{i=1}^6$ with binary labels $y^i_t \in \{0, 1\}$, the evaluated model M is required to maximize the probability of the predicated impact $\hat{Y}_t = \{\hat{y}^i_t\}_{i=1}^6$ towards ground-truth." (p.5)
Q36	"We formulate the ranking-based QA task by prompting the models to identify the likelihood of each article containing the correct answer from a candidate pool." (p.5)
Q117	"The classification task is binary (true/false), requiring the model to identify whether the text explicitly mentions any of the defined impacts." (p.16)
Q118	"The guidelines emphasize focusing on direct and immediate effects, ensuring that classifications are based solely on explicit references within the text." (p.16)
Q147	"Return 'false' for any impact category that is either not present in the text or not related to weather events. Base classifications on explicit mentions in the text. Focus on direct impacts rather than implications. Consider immediate and direct effects." (p.20)
D17	PETER MARTIN, GAZETTE Workers from the Connecticut Light and Power Co were working to reconnect power lines on Oxford Ave in Notre Dame de Grace... Temperatures plummet; today's high minus-15C... more than one million people struggle on without power or heat — 1998 Quebec ice storm coverage describing power outages and extreme cold; labeled all-zero — potential annotation gap
D27	A poor lady passenger was dashed to leeward and had her skull fractured. She was landed this morning in a dying state with a coffined child that succumbed last night to its sufferings. — 1881 Atlantic storm with explicit casualties (skull fracture, child death) labeled all-zero

Strengths

- User explicitly designed deployment to use binary labels as upstream features for downstream credibility aggregation [\[A3\]](#) — output schema is functionally compatible
- Detailed operational criteria provided for each label [\[Q141–Q146\]](#) enable consistent interpretation
- Multi-label simultaneous output [\[Q57\]](#) supports reports spanning multiple impact dimensions

- Six-category schema accepted as broadly applicable to Argentine disasters per elicitation [A1]

ID	Evidence
Q57	"Different from traditional methods that decompose multi-label text classification into multiple binary classification tasks (Boutell et al., 2004; Liu et al., 2017), we simultaneously identify all relevant disruptive weather impacts for each input by calling the LLM once." (p.6)
Q141	"Infrastructural Impact: Classify as 'true' if the text mentions any damage or disruption to physical infrastructure and essential services. This includes structural damage to buildings, roads, or bridges; any disruptions to transportation systems such as railway cancellations or road closures; interruptions to public utilities including power and water supply; any failures in communication networks; or damage to industrial facilities. Consider only explicit mentions of physical damage or service disruptions in your classification." (p.20)
Q142	"Agricultural Impact: Classify as 'true' if the text mentions any weather-related effects on farming and livestock management operations. This includes yield variations in crops and animal products; direct damage to crops, timber resources, or livestock; modifications to agricultural practices or schedules; disruptions to food production or supply chains; impacts on farming equipment and resources; or effects on agricultural inputs including soil conditions, water availability for farming, and essential materials such as seedlings, fertilizers, or animal feed." (p.20)
Q143	"Ecological Impact: Classify as 'true' if the text mentions any effects on natural environments and ecosystems. This includes alterations to local environments and biodiversity; impacts on wildlife populations and behavior patterns; effects on non-agricultural plant life and vegetation; modifications to natural habitats including water bodies, forests, and wetlands; changes in hydrological systems such as river levels and lake conditions; or impacts on urban plant life." (p.20)
Q144	"Financial Impact: Classify as 'true' if the text explicitly mentions economic consequences of weather events. This includes direct monetary losses; business disruptions or closures requiring financial intervention; market price fluctuations or demand changes for specific goods; impacts on tourism and local economic activities; or insurance claims or economic relief measures. Focus only on explicit mentions of financial losses or fluctuations." (p.20)
Q145	"Human Health Impact: Classify as 'true' if the text mentions physical or mental health effects of weather events on populations. This includes direct injuries or fatalities (including cases where zero or more casualties are explicitly mentioned); elevated risks of weather-related or secondary illnesses; mental health consequences such as stress or anxiety; impacts on healthcare service accessibility; or long-term health implications." (p.20)
Q146	"Political Impact: Classify as 'true' if the text mentions governmental and policy responses to weather events. This includes government decision-making and policy modifications in response to events; changes in public opinion or political discourse; effects on electoral processes or outcomes; international relations and aid responses; or debates surrounding disaster preparedness and response capabilities." (p.20)

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Six output labels (Infrastructural, Political, Financial, Ecological, Agricultural, Human Health) [[Q25](#)] are regionally relevant per elicitation [A1] and confirmed empirically present [[DATASET-D14](#), [D16](#), [D18](#), [D21](#)].

OO-2: Missing categories: (a) credibility / source-reliability / signal-to-noise dimension — the deployment's core decision output, addressed downstream per A3; (b) sub-labels for villa miseria infrastructure vs. formal infrastructure; (c) sub-labels for slow-onset ecological/agricultural impacts (ENSO drought) — guidelines exclude inferred consequences [[Q118](#), [Q147](#)]; (d) misinformation likelihood scoring.

OO-3: Categories themselves do not encode non-Argentine values per elicitation [A4]; however, Political Impact criterion [[Q140](#), [Q146](#)] frames responses around 'governmental and policy responses' in a Canadian institutional frame that may not capture Argentine partisan polarization or villa miseria framing. The 'direct and immediate' framing [[Q118](#), [Q147](#)] systematically

excludes the slow-onset and politically-instrumentalized framings common in Argentine disaster discourse.

OO-4: Stakeholder-driven taxonomy redesign is recommended for sub-label addition (informal settlement vulnerability, agricultural commodity-specific loss, partisan-framing political impact). User indicates failure analysis on Argentine data could inform future sub-labeling [A4].

OO-5: Structural validity is harmed by absence of credibility/reliability output (the deployment's primary need) and absence of sub-labels for Argentine-specific granularity. Empirical labeling inconsistencies [DATASET-D17, D22, D27] additionally raise concern about the structural reliability of the binary outputs.

ID	Evidence
Q25	"Six vulnerability-related disruptive weather impacts are defined as the labeling categories, including Infrastructural, Political, Financial, Ecological, Agricultural, and Human Health." (p.4)
Q117	"The classification task is binary (true/false), requiring the model to identify whether the text explicitly mentions any of the defined impacts." (p.16)
Q118	"The guidelines emphasize focusing on direct and immediate effects, ensuring that classifications are based solely on explicit references within the text." (p.16)
Q140	"Political Impact: Evaluates governmental and policy responses to weather events. Includes government decision-making and policy changes; shifts in public opinion or political discourse; effects on electoral processes; international aid and relations; and debates on disaster preparedness and response. Both direct political reactions and policy implications should be analyzed." (p.19)
Q146	"Political Impact: Classify as 'true' if the text mentions governmental and policy responses to weather events. This includes government decision-making and policy modifications in response to events; changes in public opinion or political discourse; effects on electoral processes or outcomes; international relations and aid responses; or debates surrounding disaster preparedness and response capabilities." (p.20)
Q147	"Return 'false' for any impact category that is either not present in the text or not related to weather events. Base classifications on explicit mentions in the text. Focus on direct impacts rather than implications. Consider immediate and direct effects." (p.20)
D14	by six o'clock the electric car services in all the important points west of Toronto had been completely paralyzed... an unprecedented snowfalls and heavy winds. — 1894 blizzard; infrastructure (transit) disruption labeled; North American geography
D16	Aid Laurent said that there were mountains of snow all over the city. Already \$30,000 has been expended in removing it... Seven hundred and fifty men and 450 carters were employed... — 1887 Montreal snow removal debate in city council; all three labeled labels are plausible
D17	PETER MARTIN, GAZETTE Workers from the Connecticut Light and Power Co were working to reconnect power lines on Oxford Ave in Notre Dame de Grace... Temperatures plummet; today's high minus-15C... more than one million people struggle on without power or heat — 1998 Quebec ice storm coverage describing power outages and extreme cold; labeled all-zero — potential annotation gap
D18	more than 100,000 homes damaged and crops on about 175,000 hectares of farmland destroyed... death toll in Chongqing in China's southwest rose to 42 people and 12 missing from torrential downpours — 2007 global flood roundup (England, China, Pakistan) — non-Canadian events used as positive examples
D21	hundreds of houses, schools and other buildings sustained damage when one of the twin peaks of the volcano exploded. At least 20 people sustained minor injuries... The most powerful earthquake to strike western Washington state in 30 years injured four people — 1999 global disaster roundup (Indonesia, Washington state, NZ, Romania); labeled positive
D22	Almost 4 million people across China had been cut off by flood waters, 810,000 homes have collapsed and 2.8 million homes have been damaged in eight provinces... Japan struck at young and old yesterday, killing a schoolgirl and an 85-year-old woman — 1996 China Yangtze floods + Japan E. coli outbreak in same chunk; infrastructural=1 but human_health=0 despite explicit casualties
D27	A poor lady passenger was dashed to leeward and had her skull fractured. She was landed this morning in a dying state with a cofined child that succumbed last night to its sufferings. — 1881 Atlantic storm with explicit casualties (skull fracture, child death) labeled all-zero

Information Gaps

- Whether Argentine policy users would accept 'direct and immediate' criterion or require slow-onset assessment for ENSO drought
- Specification of downstream credibility aggregation pipeline that A3 references

Requires Expert Verification

- Whether the labeling inconsistencies flagged in dataset analysis [D17, D22, D27] constitute systematic errors or chunk-level re-annotation per Q47/Q48

Remediation

Gap 1: No native credibility/reliability output and no sub-labels for Argentine-specific granularity [Q117, A3]

Recommendation: Design and document the downstream credibility aggregation layer that A3 envisions, specifying Argentine source-metadata schema (outlet reach, account verification, AFE/SINAGIR/SMN actor recognition). Add sub-labels for villa miseria infrastructure, agricultural commodity specificity (soybean/maize/wheat), and partisan-framing political impact. Treat WXImpactBench outputs as one of several upstream features.

ID	Evidence
Q117	"The classification task is binary (true/false), requiring the model to identify whether the text explicitly mentions any of the defined impacts." (p.16)

Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: WXIMPACTBENCH | Context: Argentine Government Disaster Reporting Reliability — Policy Staff

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Annotators are three domain experts from a research group 'specializing in uncovering the history of a region's climate change through the regional historical weather records' [Q110, Q111] — i.e., specialists in Canadian historical climate. No demographic information about annotator nationality, language background, or familiarity with Latin American disaster communication is documented. The paper acknowledges 'the interpretation of weather-related disruptions in historical newspapers might be influenced by demographic and contextual factors' [Q96] but does not mitigate this. The user does not anticipate systematic disagreement on formal press given Argentine reliance on Global North standards [A4], which moderates concern. However, dataset analysis surfaces concrete labeling inconsistencies: 1998 Quebec ice storm (1M+ without power, -30 °C) labeled all-zero on infrastructural and human_health [DATASET-D17]; 1881 storm with explicit fatalities labeled human_health=0 [DATASET-D27]; China flood + Japan deaths labeled infra=1 but human_health=0 [DATASET-D22] while a smaller-scale Indonesian eruption is labeled human_health=1 [DATASET-D21]. These inconsistencies raise convergent validity concerns even within the Canadian context, compounded by the absence of any Argentine validation. Output content is MODERATE-priority per elicitation; score reflects the combination of zero Argentine annotator representation and empirical labeling inconsistencies.

ID	Evidence
Q96	"The interpretation of weather-related disruptions in historical newspapers might be influenced by demographic and contextual factors, which is similar to other text datasets generated through crowd-sourcing with inherent challenges in ensuring that dataset labels are fully representative of diverse societal perspectives (Talat et al., 2022)." (p.9)
Q110	"The annotation process was conducted by members of a research group specializing in uncovering the history of a region's climate change through the regional historical weather records." (p.15)
Q111	"Their expertise can ensure the accuracy and reliability of annotations." (p.15)
D17	PETER MARTIN, GAZETTE Workers from the Connecticut Light and Power Co were working to reconnect power lines on Oxford Ave in Notre Dame de Grace... Temperatures plummet; today's high minus-15C... more than one million people struggle on without power or heat — 1998 Quebec ice storm coverage describing power outages and extreme cold; labeled all-zero — potential annotation gap
D21	hundreds of houses, schools and other buildings sustained damage when one of the twin peaks of the volcano exploded. At least 20 people sustained minor injuries... The most powerful earthquake to strike western Washington state in 30 years injured four people — 1999 global disaster roundup (Indonesia, Washington state, NZ, Romania); labeled positive
D22	Almost 4 million people across China had been cut off by flood waters, 810,000 homes have collapsed and 2.8 million homes have been damaged in eight provinces... Japan struck at young and old yesterday, killing a schoolgirl and an 85-year-old woman — 1996 China Yangtze floods + Japan E. coli outbreak in same chunk; infrastructural=1 but human_health=0 despite explicit casualties
D27	A poor lady passenger was dashed to leeward and had her skull fractured. She was landed this morning in a dying state with a cofined child that succumbed last night to its sufferings. — 1881 Atlantic storm with explicit casualties (skull fracture, child death) labeled all-zero

Strengths

- Annotation guidelines reviewed by meteorological specialists [Q104]
- Independent chunk-level re-annotation for mixed-context version [Q47, Q48] reduces label inheritance bias
- Topic-aware article selection by historical climate researchers reduces temporal/geographic selection bias [Q95]
- Detailed per-label criteria [Q141–Q146] support reproducibility

- User does not anticipate systematic label disagreement for formal press given Argentine reliance on Global North standards [A4]

ID	Evidence
Q47	"Note that annotations for these smaller chunks are performed independently by the same domain experts rather than automatically inherited from the original articles." (p.6)
Q48	"This independent annotation process naturally results in some chunks containing no weather impact labels, which serve as valuable negative examples in our evaluation." (p.6)
Q95	"Topic-aware article selection is conducted by researchers specializing in historical climate analysis to ensure the dataset is not biased on specific time and location." (p.9)
Q104	"To ensure high-quality and consistent annotations, the task was conducted using a set of specific instructions reviewed by meteorological experts." (p.15)
Q141	"Infrastructural Impact: Classify as 'true' if the text mentions any damage or disruption to physical infrastructure and essential services. This includes structural damage to buildings, roads, or bridges; any disruptions to transportation systems such as railway cancellations or road closures; interruptions to public utilities including power and water supply; any failures in communication networks; or damage to industrial facilities. Consider only explicit mentions of physical damage or service disruptions in your classification." (p.20)
Q142	"Agricultural Impact: Classify as 'true' if the text mentions any weather-related effects on farming and livestock management operations. This includes yield variations in crops and animal products; direct damage to crops, timber resources, or livestock; modifications to agricultural practices or schedules; disruptions to food production or supply chains; impacts on farming equipment and resources; or effects on agricultural inputs including soil conditions, water availability for farming, and essential materials such as seedlings, fertilizers, or animal feed." (p.20)
Q143	"Ecological Impact: Classify as 'true' if the text mentions any effects on natural environments and ecosystems. This includes alterations to local environments and biodiversity; impacts on wildlife populations and behavior patterns; effects on non-agricultural plant life and vegetation; modifications to natural habitats including water bodies, forests, and wetlands; changes in hydrological systems such as river levels and lake conditions; or impacts on urban plant life." (p.20)
Q144	"Financial Impact: Classify as 'true' if the text explicitly mentions economic consequences of weather events. This includes direct monetary losses; business disruptions or closures requiring financial intervention; market price fluctuations or demand changes for specific goods; impacts on tourism and local economic activities; or insurance claims or economic relief measures. Focus only on explicit mentions of financial losses or fluctuations." (p.20)
Q145	"Human Health Impact: Classify as 'true' if the text mentions physical or mental health effects of weather events on populations. This includes direct injuries or fatalities (including cases where zero or more casualties are explicitly mentioned); elevated risks of weather-related or secondary illnesses; mental health consequences such as stress or anxiety; impacts on healthcare service accessibility; or long-term health implications." (p.20)
Q146	"Political Impact: Classify as 'true' if the text mentions governmental and policy responses to weather events. This includes government decision-making and policy modifications in response to events; changes in public opinion or political discourse; effects on electoral processes or outcomes; international relations and aid responses; or debates surrounding disaster preparedness and response capabilities." (p.20)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Ground-truth labels reflect Canadian historical-climate research-group perspectives [Q110, Q111]; Argentine stakeholder perspectives are not represented in any annotation step.

OC-2: Elicitation suggests minimal expected disagreement on formal press [A4] but acknowledges fluidity for informal/politically charged Argentine content. Empirical labeling inconsistencies [DATASET-D17, D22, D27] suggest within-Canadian-context label noise that compounds any cross-cultural transfer concern.

OC-3: INSUFFICIENT DOCUMENTATION on annotator demographics: paper specifies only research-group affiliation and meteorological expertise [Q110, Q111, Q104, Q105]; no nationality, language, age, gender, or cultural-context disclosure. No Datasheet or Data Statement equivalent provided.

OC-4: Re-annotation by representative Argentine annotator pool would be necessary to validate transfer; no Argentine Spanish disaster benchmark exists [WEB-14, gap_id 7] to substitute.

OC-5: Aggregation method is not documented (whether single-annotator-per-chunk, majority vote, or adjudication); paper says 'three domain annotators following our guidelines' [Q26] but does not specify inter-annotator agreement or aggregation procedure — INSUFFICIENT DOCUMENTATION on minority-perspective preservation.

OC-6: Convergent validity is at risk: zero Argentine annotator representation, plus empirical labeling inconsistencies on chunks containing explicit casualties or major utility outages [DATASET-D17, D22, D27]. External validity to Argentine context cannot be established without re-annotation.

ID	Evidence
Q26	"Annotation is conducted by three domain annotators following our guidelines (provided in Appendix B.2.2)." (p.4)
Q96	"The interpretation of weather-related disruptions in historical newspapers might be influenced by demographic and contextual factors, which is similar to other text datasets generated through crowd-sourcing with inherent challenges in ensuring that dataset labels are fully representative of diverse societal perspectives (Talat et al., 2022)." (p.9)
Q104	"To ensure high-quality and consistent annotations, the task was conducted using a set of specific instructions reviewed by meteorological experts." (p.15)
Q105	"The annotation guideline and the categories definition are provided in Table 14 and Table 15, respectively." (p.15)
Q110	"The annotation process was conducted by members of a research group specializing in uncovering the history of a region's climate change through the regional historical weather records." (p.15)
Q111	"Their expertise can ensure the accuracy and reliability of annotations." (p.15)
D17	PETER MARTIN, GAZETTE Workers from the Connecticut Light and Power Co were working to reconnect power lines on Oxford Ave in Notre Dame de Grace... Temperatures plummet; today's high minus-15C... more than one million people struggle on without power or heat — 1998 Quebec ice storm coverage describing power outages and extreme cold; labeled all-zero — potential annotation gap
D22	Almost 4 million people across China had been cut off by flood waters, 810,000 homes have collapsed and 2.8 million homes have been damaged in eight provinces... Japan struck at young and old yesterday, killing a schoolgirl and an 85-year-old woman — 1996 China Yangtze floods + Japan E. coli outbreak in same chunk; infrastructural=1 but human_health=0 despite explicit casualties
D27	A poor lady passenger was dashed to leeward and had her skull fractured. She was landed this morning in a dying state with a confined child that succumbed last night to its sufferings. — 1881 Atlantic storm with explicit casualties (skull fracture, child death) labeled all-zero
WEB-14	No complementary Argentine Spanish-language benchmark exists for cross-validation (https://crisisnlp.qcri.org/humaid_dataset)

Information Gaps

- Annotator nationality, language, and cultural-context demographics
- Inter-annotator agreement statistics and aggregation method
- Whether observed labeling inconsistencies [D17, D22, D27] reflect chunk-level re-annotation policy or genuine errors

Requires Expert Verification

- Whether Argentine emergency management officials would assign positive labels to chunks like D17 and D27 under the same guideline criteria

Remediation

Gap 1: Zero Argentine annotator representation; empirical labeling inconsistencies on positively-framed chunks [Q110, DATASET-D17, D22, D27]

Recommendation: Re-annotate a stratified sample (≥ 100 chunks) including the flagged inconsistencies (D17 ice storm, D22 China/Japan, D27 1881 storm) using ≥ 3 Argentine emergency management experts. Compute inter-annotator agreement against original Canadian labels to quantify cross-cultural drift, and use disagreements to inform Argentine-specific sub-label development per A4.

ID	Evidence
Q110	"The annotation process was conducted by members of a research group specializing in uncovering the history of a region's climate change through the regional historical weather records." (p.15)
D17	PETER MARTIN, GAZETTE Workers from the Connecticut Light and Power Co were working to reconnect power lines on Oxford Ave in Notre Dame de Grace... Temperatures plummet; today's high minus-15C... more than one million people struggle on without power or heat — 1998 Quebec ice storm coverage describing power outages and extreme cold; labeled all-zero — potential annotation gap
D22	Almost 4 million people across China had been cut off by flood waters, 810,000 homes have collapsed and 2.8 million homes have been damaged in eight provinces... Japan struck at young and old yesterday, killing a schoolgirl and an 85-year-old woman — 1996 China Yangtze floods + Japan E. coli outbreak in same chunk; infrastructural=1 but human_health=0 despite explicit casualties
D27	A poor lady passenger was dashed to leeward and had her skull fractured. She was landed this morning in a dying state with a coffined child that succumbed last night to its sufferings. — 1881 Atlantic storm with explicit casualties (skull fracture, child death) labeled all-zero

Output Form (OF)

Score: 3/5 — Moderate gaps | Confidence: medium

Benchmark: WXIMPACTBENCH | Context: Argentine Government Disaster Reporting Reliability — Policy Staff

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output form consists of binary label vectors and ranked article lists [Q33, Q36, Q43–Q45, Q50], producing F1, accuracy, row-wise accuracy, Hit@1, nDCG@5, Recall@5, and MRR. These outputs are usable as upstream features for the deployment's downstream credibility aggregation per A3, so the form is functionally compatible. However, the benchmark does not natively produce the credibility/reliability score that constitutes the deployment's primary decision output, and the transformation from binary labels to credibility scores requires non-trivial integration with Argentine source metadata (outlet reach, account verification, institutional affiliation) that is unaddressed. Row-wise accuracy [Q45, Q75] provides a stricter evaluation that more closely approximates aggregation reliability needs. Output form is MODERATE-priority per elicitation; score reflects functional compatibility tempered by the non-trivial transformation gap and the absence of confidence/probability outputs that would be most useful for downstream aggregation.

ID	Evidence
Q33	"Specifically, given an article sample x_t corresponding to six ground-truth impacts $Y_t = \{y^i_t\}_{i=1}^6$ with binary labels $y^i_t \in \{0, 1\}$, the evaluated model M is required to maximize the probability of the predicated impact $\hat{Y}_t = \{\hat{y}^i_t\}_{i=1}^6$ towards ground-truth." (p.5)
Q36	"We formulate the ranking-based QA task by prompting the models to identify the likelihood of each article containing the correct answer from a candidate pool." (p.5)
Q43	"For multi-label classification task, we use F1-score, accuracy, and row-wise accuracy as evaluation metrics." (p.6)
Q44	"The evaluation via F1-score and accuracy are averaged across the six impact categories, historical and modern articles, and the effect of different context lengths." (p.6)
Q45	"Compared to the common F1-score and accuracy, the row-wise accuracy is a strict metric that requires more accurate output as the model should correctly classify all six impact labels for a given article." (p.6)
Q50	"For the ranking-based QA task, we deploy the standard metric that emphasizes the accuracy of top positions for evaluation, including Hit@1, nDCG@5, Recall@5, and MRR." (p.6)
Q75	"Row-wise accuracy requires an exact match across all labels." (p.8)

Strengths

- Binary label vectors and ranked lists are functionally compatible as upstream features for credibility aggregation per A3
- Multiple metrics including stricter row-wise accuracy [Q45] capture different aspects of reliability
- Standard IR metrics (Hit@1, nDCG@5, Recall@5, MRR) [Q50] for the QA task align with retrieval-pipeline evaluation
- Modest computational requirements [Q127, Q128] enable replication on Argentine-relevant content

ID	Evidence
Q45	"Compared to the common F1-score and accuracy, the row-wise accuracy is a strict metric that requires more accurate output as the model should correctly classify all six impact labels for a given article." (p.6)
Q50	"For the ranking-based QA task, we deploy the standard metric that emphasizes the accuracy of top positions for evaluation, including Hit@1, nDCG@5, Recall@5, and MRR." (p.6)
Q127	"For all open-source models, evaluations were performed on a system with two NVIDIA A6000 (32GB) GPUs." (p.17)
Q128	"The relatively modest computational requirements highlight the accessibility of our benchmark for researchers with limited computational resources, while still enabling comprehensive evaluation of state-of-the-art models" (p.17)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Output modality (binary labels + rankings) is text-based and compatible with deployment's text channels; functional for upstream feature use [A3]. However, native credibility/reliability score is absent — the user accepts this and plans downstream aggregation [A3].

OF-2: Text-to-speech is not relevant: target users are policy professionals consuming text-based reports [target_population description]; literacy is near-universal (~99%) [WEB-2, WEB-3] and not a constraint.

OF-3: Literacy and education are not limiting factors (target population holds advanced degrees; national literacy ~99% [WEB-2, WEB-3, WEB-4]). Output form's text-only nature is appropriate.

OF-4: External validity gap: benchmark does not produce probability scores or confidence outputs that would be most useful for downstream credibility aggregation; only binary labels and discrete rankings. Integration with Argentine source metadata (outlet, account verification, institutional affiliation) is not addressed within benchmark scope.

ID	Evidence
Q33	"Specifically, given an article sample x_t corresponding to six ground-truth impacts $Y_t = \{y^i_t\}_{i=1}^6$ with binary labels $y^i_t \in \{0, 1\}$, the evaluated model M is required to maximize the probability of the predicated impact $\hat{Y}_t = \{\hat{y}^i_t\}_{i=1}^6$ towards ground-truth." (p.5)
Q43	"For multi-label classification task, we use F1-score, accuracy, and row-wise accuracy as evaluation metrics." (p.6)
WEB-2	Argentina national adult literacy ~99% (https://countryeconomy.com/demography/literacy-rate/argentina)
WEB-3	World Bank/UNESCO confirms literacy >98% since 2001 (https://www.macrotrends.net/global-metrics/countries/arg/argentina/literacy-rate)
WEB-4	OECD 2025: 19% tertiary qualification among 25–34 year-olds; target cohort holds advanced degrees (https://gpseducation.oecd.org/CountryProfile?primaryCountry=ARG&treshold=10&topic=EO)

Information Gaps

- Whether models can produce calibrated probability outputs in addition to binary labels for downstream credibility aggregation
- Specification of the Argentine source-metadata schema for credibility aggregation

Requires Expert Verification

- Design of the credibility aggregation layer that combines benchmark outputs with Argentine source metadata

Remediation

Gap 1: Binary labels only; no calibrated probability outputs for downstream credibility aggregation [Q33]

Recommendation: Re-run benchmark evaluation extracting model log-probabilities or calibrated confidence scores per label rather than binary outputs. Evaluate whether confidence calibration on WXImpactBench correlates with calibration on the Argentine test set built under remediation 1, to inform the credibility aggregation layer.

ID	Evidence
Q33	“Specifically, given an article sample x_t corresponding to six ground-truth impacts $Y_t = \{y^i_t\}_{i=1}^6$ with binary labels $y^i_t \in \{0, 1\}$, the evaluated model M is required to maximize the probability of the predicated impact $\hat{Y}_t = \{\hat{y}^i_t\}_{i=1}^6$ towards ground-truth.” (p.5)